Will Gregory

william-gregory.github.io
william.gregory@ucl.ac.uk
github.com/William-gregory
linkedin.com/in/will-gregory
scholar.google.com

Education

2017-2021	PhD in Polar Climate Science and ML, *Pass (no corrections)*	University College London (UCL)
2013-2014	MSc in Petroleum Geophysics, *Distinction*	Imperial College London
2010-2013	BSc in Geology with Geophysics, *1st class*	University of Leicester

Employment

06/2026-present **Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellow | UCL**
Project: "Data Assimilation and Machine learning for online Parameter Estimation (DAMPEn)", aims to utilise large-scale remote sensing observations to learn state-dependent sea ice model physics parameters. In collaboration with the UK Met Office and the University of Tromsø.

01/2022-05/2026 **Postdoctoral Researcher | Princeton University & Geophysical Fluid Dynamics Lab**
Funded by Multi-scale Machine Learning In coupled Earth System modeling (M^2LInES) initiative
Past Responsibilities:
- Lead coupled ocean-sea-ice emulation development through M^2LInES–AI2 partnership
- Lead hybrid sea ice modelling development in M^2LInES and GFDL
- Lead organiser of the 2025 M^2LInES annual meeting
- Co-organiser of biweekly project meetings

11/2014-07/2017 **Depth-imaging Geophysicist | Petroleum Geo-Services Ltd**
Client-facing role in the private sector, delivering geophysical services to oil and gas companies
Past Responsibilities:
- Develop sub-surface seismic velocity models using tomographic inversion techniques
- Deliver project presentations and lead data visualisation sessions with clients
- Develop and maintain Gantt charts, status reports, and end-of-project reports
- Train new staff in sub-surface imaging theory and in-house software

Successful and Pending Research Funding

Royal Society University Research Fellowship (2027)	PI; I designed and authored a full 8-year independent research programme on AI-based coupled climate emulation and prediction systems, including tailored projects and training and development plans for two PhD students. *Pending*
Marie Curie Postdoctoral Fellowship (2026)	I designed and authored a full 2-year independent research programme on AI for climate model parameterisation, including scientific objectives, methodology, personal and professional development, and outreach and dissemination. *Successful, 98%*

Service

2026	Invited as expert panelist to an ECMWF workshop on "Towards coupled AI climate models".
2026	Lead convener on a proposed session related to sea ice modelling at EGU 2027.
2026	Guest Editor for a special edition, "AI-driven Ocean Science", in BMC Marine Science (Springer Nature).
2026	Invited to discuss climate modelling and AI with the UN Secretary-General's Scientific Advisory Board.
2026	EGU General Assembly Outstanding Student and PhD candidate Presentation (OSPP) judge.
2025	Invited panelist to the Science Philanthropy Alliance meeting, "The Next Generation".
2024 & 2025	Proposal reviewer for the National Science Foundation (NSF) Arctic Natural Sciences program.
2024 & 2026	AGU Annual Meeting and Ocean Sciences Outstanding Student Presentation Award (OSPA) judge.
2024	AGU Annual Meeting convener and chair for session: "Data Driven Science: Developments in Machine Learning Subgrid-Scale Parameterizations and in Reanalyses across Earth System Modeling".
2019-present	Peer reviewer for numerous international journals, including Science Advances, Geophysical Research Letters, npj Climate and Atmospheric Science, and more.

Selected Honours

Fellowships

2026-2028	Marie Skłodowska-Curie Actions (MSCA) fellowship | Awarded €260,347 for 2-year postdoctoral fellowship
2017-2021	London NERC Doctoral Training Partnership (DTP) | Funding for 4-yr PhD, amounting to over £91,000

Awards

2025	Wiley top viewed article award | Gregory et al in top 10% of most-viewed 2023 papers in *JAMES*
2013	British Petroleum Scholarship, Imperial College London | £25,000 to study MSc Petroleum Geophysics
2013	Shell Geophysics Prize | Awarded for highest contributions to BSc Geophysics programme

Talks and Seminars

Select invited talks

"Machine learning for sea ice prediction: from where have we come and to where are we going?", AGU Annual Meeting, San Francisco, USA (2026)

"A new global sea ice emulator and the path toward fully data-driven coupled climate modeling", AGU Annual Meeting, San Francisco, USA (2026)

"A new global sea ice emulator and the path toward fully data-driven coupled climate modelling", ORCAS Community Meeting (Observational Requirements in the Context of AI Prediction Systems for Sea Ice), Virtual (2026)

"How is AI shaping sea ice modeling and prediction research?", Polar Research Board, Fall Meeting, Virtual (2025)

"How is AI shaping Arctic climate modeling and observational research?", US Interagency Arctic Research Policy Committee (IARPC) Webinar, Virtual (2025)

"Bridging AI, remote sensing, and climate modeling to understand polar climate variability and change", Institute for Advanced Computational Science (IACS) Seminar, Stony Brook University, USA (2025)

"From component to coupled: evaluating the performance of a machine-learned sea ice bias correction scheme in fully-coupled seasonal predictions", Nansen SuperIce Webinar Series, Virtual (2024)

"Applications of machine learning to sea ice data assimilation", 10th US Climate modeling summit, GFDL, Princeton, USA (2024)

"Towards a machine-learned sea ice model parameterization from data assimilation increments", Euro-Mediterranean Center on Climate Change (CMCC) Seminar, Bologna, Italy (2024)

"Deep learning of systematic sea ice model errors from data assimilation increments", Courant Institute of Mathematical Sciences Guest Seminar Series, New York City, USA (2022)

"Machine learning tools for pattern recognition in polar climate science", EGU General Assembly, Vienna, Austria (2021)

Recent conference talks

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "Integrating an interactive sea ice component into an AI-based coupled climate emulation framework," UK Sea Ice Meeting, Reading, UK (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "Towards a mass-conservative global sea ice emulator that generalizes across climates," EGU General Assembly, Vienna, Austria (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "Towards a mass-conservative global sea ice emulator that generalizes across climates," Climate Informatics, Lausanne, Switzerland (2026)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna, C McHugh, L Jia. "Evaluating a new hybrid sea ice model in global fully-coupled climate predictions and projections," AGU Ocean Sciences Meeting, Glasgow, UK (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, S Clark, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "FloeNet: a mass-conserving emulator of the GFDL sea ice model," CESM Polar Working Group Meeting, Boulder, USA (2026)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. "Machine-leaned sea ice bias correction in a global fully-coupled climate model," Gordon Research Conference: Machine Learning for Actionable Climate Science, RI (2025)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. "Machine-leaned sea ice bias correction in a global fully-coupled climate model," CESM Polar Working Group Meeting, Boulder, CO (2025)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. "Towards improving numerical sea ice predictions with data assimilation and machine learning," Cross-VESRI Convening, Cambridge, UK (2024)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. "Towards improving numerical sea ice predictions with data assimilation and machine learning," EGU General Assembly, Vienna, Austria (2024)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. "Machine learning for online sea ice bias correction within global ice-ocean simulations, AGU Ocean Sciences Meeting, New Orleans, USA (2024)

Teaching & Mentoring

2020	TA in BSc Geodynamics, UCL | Class size: 15-20. Format: Tutorials
2020	TA in BSc Ocean Physics, UCL | Class size: 15. Format: Tutorials
2019	TA in BSc Principles of Climate, UCL | Class size: 10-15. Format: Tutorials
2018-2019	TA in BSc Foundations of Physical Geoscience, UCL | Class size: 10-12. Format: Lectures and Tutorials
2018-2020	TA in BSc Introduction to Matlab, UCL | Class size: 15-20. Format: Lectures and Tutorials
2026-2027	Designed an MSci project where I will be direct supervisor of UCL Earth Sciences MSci student, Simi Rayat. The project involves using dual altimetry missions, SWOT and ICESat-2, to investigate the physical consistency of regional short term sea ice thickness variability within machine-learning-derived (Gaussian Processes, Deep Random Features) data products.
2022-2023	With Prof. Marc Deisinroth, I co-supervised Ronald MacEachern, a postgraduate student undertaking the MSc Machine Learning course within the Department of Computer Science at UCL. I designed the MSc dissertation project, which was titled "Sea Ice Freeboard Interpolation using Gaussian Process Regression". Ronald achieved a thesis grade of Distinction. I then led an international collaboration to build Ronald's work into an open-source Python library, GPSat, along with the *Nature Communications* paper Gregory, MacEachern, Takao et al 2024.

Public Engagement & Outreach

2023	"AI is transforming climate forecasts for melting sea ice," *Advanced Science News Article*
2023-2024	Project development chair for the Climatematch academy outreach program | Independently managed 4 individuals who were developing student project materials
2022-2023	Curriculum content reviewer for the Climatematch Academy outreach programme | Responsible for reviewing all content relating to fundamentals of climate science
2022	Presented on EDI progress within NOAA and GFDL, Princeton University and GFDL, NJ
2016	Delivered science outreach presentations to years 10-12, King Solomon Academy school, London UK
2012-2013	Assisted in the Department of Geology open days at the University of Leicester | Responsible for welcoming members of the public to the department and also talking to prospective students about the Geology with Geophysics BSc programme